

SATHWIK H NAIK

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PROFESSIONAL SUMMARY

AI engineer with hands-on Python experience building and evaluating RAG systems using ChromaDB, Ollama, and RAGAS. Completed M.S. in Data Science at UMD in 2026, backed by a hybrid-search retrieval ablation and GAN-based generative research.

EDUCATION

University of Maryland, College Park

2024 – 2026

M.S. in Data Science

GPA 3.81/4.0

Coursework Statistical Inference, Deep Learning, Data Modeling, Data Visualization, Experimental Design, Advanced Time Series

Reva University, India

2019 – 2023

B.Tech, Computer Science and Engineering

GPA 3.55/4.0

WORK EXPERIENCE

Elevance Health

Aug 2023 – Aug 2024

Software Engineer

- Automated recurring data-preparation and reporting workflows, eliminating 20+ hours of manual weekly effort
- Developed ETL workflows to source, clean, and validate healthcare data, improving quality for stakeholder reports
- Designed reusable KPI structures, translating operational requirements into consistent, self-serve cross-team metrics
- Built and maintained 10+ Power BI dashboards with DAX measures for multiple internal teams
- Documented report logic, metric definitions, and validation rules enabling governed self-service reporting

PROJECTS

RAG Evaluation Lab (“Retrieval Quality Lab”) ([GitHub](#))

Tech Stack: Python, ChromaDB, Ollama, RAGAS, LangChain, BM25, sentence-transformers

- Built a RAG evaluation pipeline comparing BM25, dense, hybrid, and reranked retrieval methods
- Achieved 85.3% Recall@1 with hybrid retrieval plus cross-encoder reranking, up from 69.6% BM25
- Built a 5-tab Streamlit dashboard visualizing retrieval ablations, generation quality, and latency

TinyACE: Bounded Working-Memory Context Engineering for SLMs ([GitHub](#))

Tech Stack: Python, PyTorch, TinyLlama, Phi-3 Mini, Mistral, sentence-transformers

- Developed TINYACE, a bounded working-memory framework capping context at 512 tokens for small models
- Improved Phi-3 Mini accuracy from 74% to 78% across a 50-question SciQ benchmark
- Identified failure tracking as the most critical component through systematic ablation studies

Conditional Generative Data Augmentation for Image Classification ([GitHub](#))

Tech Stack: Python, PyTorch, Conditional GAN, Conditional VAE-GAN, CNN

- Developed CGAN and Conditional VAE-GAN pipelines in PyTorch across MNIST and Fashion-MNIST
- Improved Fashion-MNIST accuracy from 63.6% to 70.8% using only 100 real samples
- Compared cGAN against a Conditional VAE-GAN hybrid across two datasets and twelve experiments

TECHNICAL SKILLS

- LLM & RAG Engineering** RAG, Hybrid Search (BM25 + Dense + RRF), Cross-Encoder Reranking, ChromaDB, Ollama, RAGAS, LangChain
- ML & AI** PyTorch, scikit-learn, Gradient Boosting, LightGBM, SHAP, GANs, VAE-GAN, CNNs, Deep Learning
- Programming & Tools** Python, pandas, R, Git, Docker, FastAPI, CI/CD
- Statistics & Methods** Statistical Inference, Retrieval/Generation Evaluation (Recall@k, MRR, EM, F1), Ablation Studies
- AI-Assisted Development** GitHub Copilot, Claude AI, Cursor, AI Debugging